Practice Problems Midterm 2

Solutions

1. (Hotel questionnaire)

   a) By iterated expectation

$$\begin{aligned} \mathrm{E}[s] &= p_{\tilde{r}}(0)\mu_{\tilde{s}\,|\,\tilde{r}}(0) + p_{\tilde{r}}(1)\mu_{\tilde{s}\,|\,\tilde{r}}(1) && (1)\\ &= 0.8 \cdot 4 + 0.2 \cdot 2 = 3.6. && (2) \end{aligned}$$

   b) Given the conditional independence assumptions, the conditional pmf of $\tilde{s}$ given $\tilde{q} = 1$ equals

$$\begin{aligned} p_{\tilde{s}\,|\,\tilde{q}}(s\,|\,1) &= \frac{p_{\tilde{s},\tilde{q}}(s,1)}{p_{\tilde{q}}(1)} && (3)\\ &= \frac{\sum_{r=0}^{1} p_{\tilde{r},\tilde{s},\tilde{q}}(r,s,1)}{\sum_{r=0}^{1} p_{\tilde{r},\tilde{q}}(r,1)} && (4)\\ &= \frac{\sum_{r=0}^{1} p_{\tilde{r}}(r)p_{\tilde{q}\,|\,\tilde{r}}(q\,|\,1)p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,r)}{\sum_{r=0}^{1} p_{\tilde{r}}(r)p_{\tilde{q}\,|\,\tilde{r}}(q\,|\,1)} && (5)\\ &= \frac{0.8 \cdot 0.25 \cdot p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,0) + 0.2 \cdot 0.9 \cdot p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,1)}{0.8 \cdot 0.25 + 0.2 \cdot 0.9} && (6)\\ &= \frac{0.2p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,0) + 0.18p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,1)}{0.38}. && (7) \end{aligned}$$

   The conditional mean given $\tilde{q} = 1$ therefore equals

$$\begin{aligned} \mu_{\tilde{s}\,|\,\tilde{q}}(1) &= \sum_{s=1}^{5} s p_{\tilde{s}\,|\,\tilde{q}}(s\,|\,1) && (8)\\ &= \sum_{s=1}^{5} s \left(\frac{0.2p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,0) + 0.18p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,1)}{0.38} \right) && (9)\\ &= \frac{0.2}{0.38} \sum_{s=1}^{5} s p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,0) + \frac{0.18}{0.38} \sum_{s=1}^{5} s p_{\tilde{s}\,|\,\tilde{r}}(s\,|\,1) && (10)\\ &= \frac{0.2}{0.38}\mu_{\tilde{s}\,|\,\tilde{r}}(0) + \frac{0.18}{0.38}\mu_{\tilde{s}\,|\,\tilde{r}}(1) && (11)\\ &= \frac{0.2 \cdot 4}{0.38} + \frac{0.18 \cdot 2}{0.38} && (12)\\ &= 3.05. && (13) \end{aligned}$$

   The observed mean is 0.55 less than the true mean.

2. (Faulty database)

a) Let $\tilde{c}$ be a random variable that represents whether the entry is corrupted, we have

$$\tilde{b} = \begin{cases} -x & \text{if } \tilde{c} = -1, \\ \tilde{a} & \text{if } \tilde{c} = 0, \\ x & \text{if } \tilde{c} = 1. \end{cases} \tag{14}$$

The conditional mean function of $\tilde{b}$ given $\tilde{c} = c$ equals

$$\mu_{\tilde{b} \mid \tilde{c}}(c) = \begin{cases} -x & \text{if } c = -1, \\ 0 & \text{if } c = 0, \\ x & \text{if } c = 1, \end{cases} \tag{15}$$

so by iterated expectation

$$\begin{aligned} \mathrm{E}[\tilde{b}] &= \mathrm{E}\left[\mu_{\tilde{b} \mid \tilde{c}}(\tilde{c})\right] && (16) \\ &= 0.1(-x) + 0.1x = 0. && (17) \end{aligned}$$

The conditional mean function of $\tilde{b}^2$ given $\tilde{c} = c$ equals

$$\mu_{\tilde{b}^2 \mid \tilde{c}}(c) = \begin{cases} x^2 & \text{if } c = -1, \\ \sigma^2 & \text{if } c = 0, \\ x^2 & \text{if } c = 1, \end{cases} \tag{18}$$

so by iterated expectation

$$\begin{aligned} \mathrm{E}[\tilde{b}^2] &= \mathrm{E}\left[\mu_{\tilde{b}^2 \mid \tilde{c}}(\tilde{c})\right] && (19) \\ &= 0.2x^2 + 0.8\sigma^2. && (20) \end{aligned}$$

We conclude that $\mathrm{Var}[\tilde{b}] = \mathrm{E}[\tilde{b}^2] - \mathrm{E}[\tilde{b}]^2 = 0.2x^2 + 0.8\sigma^2$.

3. (Weight-loss supplement)

a) The unadjusted ATE equals

$$\begin{aligned} &\text{unadjusted ATE} \\ &= \frac{0.5 - 4.6 - 7.5 + 1.1 - 3.4 - 1.4 + 3.2 - 1.9 - 3.6 + 2.4 + 1.2 + 4.2 - 2.5 - 3.5 + 4.7}{15} \\ &\quad - \frac{1.6 - 6.7 + 2.3 - 4.5 + 3.4 - 0.6 + 2.3 + 1.5 + 1.7 + 2.6 + 0.5 + 1.2 - 3.6 + 2.1 - 1.2 + 0.4}{16} \\ &= -0.93. \end{aligned} \tag{21}$$

b) We model the treatment as a random variable $\tilde{t}$, the weight loss as a random variable $\tilde{w}$ and the exercise as a random variable $\tilde{x}$. In order to adjust for the effect of exercise,

we compute

$$p_{\tilde{x}}(1) = \frac{13}{31} = 0.42, \tag{22}$$

$$\mu_{\tilde{w}\,|\,\tilde{t},\tilde{x}}(0,0) = \frac{1.6 + 2.3 - 0.6 + 2.3 + 1.5 + 1.7 + 2.6 + 0.5 + 1.2 + 2.1 - 1.2 + 0.4}{12} \tag{23}$$

$$= 1.2, \tag{24}$$

$$\mu_{\tilde{w}\,|\,\tilde{t},\tilde{x}}(0,1) = \frac{-6.7 - 4.5 + 3.4 - 3.6}{4} = -2.85, \tag{25}$$

$$\mu_{\tilde{w}\,|\,\tilde{t},\tilde{x}}(1,0) = \frac{0.5 + 1.1 - 1.9 + 2.4 + 1.2 + 4.7}{6} = 1.33, \tag{26}$$

$$\mu_{\tilde{w}\,|\,\tilde{t},\tilde{x}}(1,1) = \frac{-4.6 - 7.5 - 3.4 - 1.4 + 3.2 - 3.6 + 4.2 - 2.5 - 3.5}{9} = -2.12, \tag{27}$$

The adjusted ATE equals

$$\text{adjusted ATE} = \sum_{x=0}^{1} p_{\tilde{x}}(x)\mu_{\tilde{w}\,|\,\tilde{t},\tilde{x}}(1,x) - \sum_{x=0}^{1} p_{\tilde{x}}(x)\mu_{\tilde{w}\,|\,\tilde{t},\tilde{x}}(0,x) \tag{28}$$

$$= (0.58 \cdot 1.33 - 0.42 \cdot 2.12) - (0.58 \cdot 1.2 - 0.42 \cdot 2.85) = 0.382. \tag{29}$$

This approximates the true ATE if the potential outcomes of the weight gain/loss for treatment and control are conditionally independent of the treatment given the exercise.

4. (Three variables) Yes, it is possible. Let $\tilde{a}$, $\tilde{b}$ and $\tilde{c}$ be random variables with zero mean and unit variance, which are pairwise uncorrelated. We define:

$$\tilde{w} := \tilde{a} + \alpha\tilde{b}, \tag{30}$$

$$\tilde{y} := \tilde{b} + \beta\tilde{c}, \tag{31}$$

$$\tilde{z} := \gamma\tilde{a} + \tilde{c}, \tag{32}$$

where α, β and γ are real-valued constants. By linearity of expectation

$$\mathrm{E}\left[\tilde{w}\right] = \mathrm{E}\left[\tilde{a}\right] + \alpha\mathrm{E}[\tilde{b}] = 0, \tag{33}$$

$$\mathrm{E}\left[\tilde{y}\right] = \mathrm{E}[\tilde{b}] + \beta\mathrm{E}\left[\tilde{c}\right] = 0, \tag{34}$$

$$\mathrm{E}\left[\tilde{z}\right] = \gamma\mathrm{E}\left[\tilde{a}\right] + \mathrm{E}\left[\tilde{c}\right] = 0. \tag{35}$$

Since $\tilde{a}$, $\tilde{b}$ and $\tilde{c}$ have zero mean and are pairwise uncorrelated,

$$\mathrm{E}[\tilde{a}\tilde{b}] = \mathrm{E}\left[\tilde{a}\tilde{c}\right] = \mathrm{E}[\tilde{b}\tilde{c}] = 0. \tag{36}$$

Consequently, again by linearity of expectation,

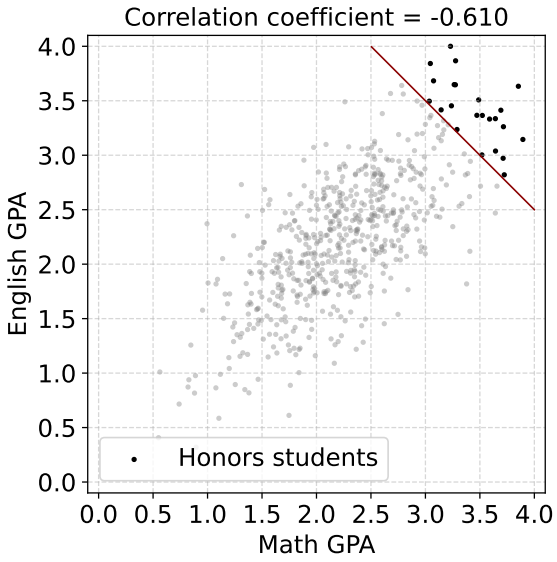
$$\begin{aligned}
\mathrm{E}\left[\tilde{w}\tilde{y}\right] &= \mathrm{E}[\tilde{a}\tilde{b}] + \beta\mathrm{E}\left[\tilde{a}\tilde{c}\right] + \alpha\mathrm{E}[\tilde{b}^2] + \alpha\beta\mathrm{E}[\tilde{b}\tilde{c}] && (37)\\
&= \alpha, && (38)\\
\mathrm{E}\left[\tilde{w}\tilde{z}\right] &= \gamma\mathrm{E}\left[\tilde{a}^2\right] + \mathrm{E}\left[\tilde{a}\tilde{c}\right] + \gamma\alpha\mathrm{E}[\tilde{a}\tilde{b}] + \alpha\mathrm{E}[\tilde{b}\tilde{c}] && (39)\\
&= \gamma, && (40)\\
\mathrm{E}\left[\tilde{y}\tilde{z}\right] &= \gamma\mathrm{E}[\tilde{a}\tilde{b}] + \mathrm{E}[\tilde{b}\tilde{c}] + \gamma\beta\mathrm{E}\left[\tilde{a}\tilde{c}\right] + \beta\mathrm{E}\left[\tilde{c}^2\right] && (41)\\
&= \beta. && (42)
\end{aligned}$$

If we choose α and β to be positive and γ negative, then $\tilde{w}$ and $\tilde{y}$ are positively correlated, and so are $\tilde{y}$ and $\tilde{z}$, but $\tilde{w}$ and $\tilde{y}$ are negatively correlated.

5. (Berkson's paradox)

   (a) The plot shows that English and Math GPAs are approximately proportional, so they are positively correlated.

   (b) Points such that the two entries sum 6.5 or more have a clear negative trend on average, as shown in the following plot. Consequently, conditioned on being an honors student, English and Math GPAs are negatively correlated.



6. (Noisy measurement)

a) We have

$$\begin{aligned}
\mathrm{Var}[\tilde{y}] &= \mathrm{Var}[\tilde{x}] + \mathrm{Var}[\tilde{z}] && (43)\\
&= 1 + \sigma^2, && (44)\\
\mathrm{E}[\tilde{x}\tilde{y}] &= \mathrm{E}[\tilde{x}^2] + \mathrm{E}[\tilde{x}\tilde{z}] && (45)\\
&= 1, && (46)\\
\rho_{\tilde{x},\tilde{y}} &= \frac{\mathrm{E}[\tilde{x}\tilde{y}]}{\sqrt{1+\sigma^2}} && (47)\\
&= \frac{1}{\sqrt{1+\sigma^2}}. && (48)
\end{aligned}$$

The linear MMSE estimate of $\tilde{x}$ given $\tilde{y} = y$ equals

$$\begin{aligned}
\ell_{\mathrm{MMSE}}(y) &= \frac{\rho_{\tilde{x},\tilde{y}}\sigma_{\tilde{x}} y}{\sigma_{\tilde{y}}} && (49)\\
&= \frac{y}{1+\sigma^2}. && (50)
\end{aligned}$$

b)

$$\begin{aligned}
\mathrm{E}\left[(\tilde{x} - \ell_{\mathrm{MMSE}}(\tilde{y}))^2\right] &= (1-\rho^2_{\tilde{x},\tilde{y}})\sigma^2_{\tilde{x}} && (51)\\
&= \frac{\sigma^2}{1+\sigma^2}. && (52)
\end{aligned}$$

c) If $\sigma \to 0$ the estimate equals y and the error is zero, which makes sense since in that case the noise is zero and hence $\tilde{y} = \tilde{x}$ with probability one.

d) If $\sigma \to \infty$ the estimate equals zero and the error is 1, which makes sense since in that case we are better off just ignoring the measurement and setting the estimate equal to the mean of $\tilde{x}$. The corresponding error is consequently the variance of $\tilde{x}$.

7. (Bernoulli random variables) Let $\tilde{a}$ and $\tilde{b}$ be two uncorrelated Bernoulli random variables with parameters $\theta_{\tilde{a}}$ and $\theta_{\tilde{b}}$ defined in the same probability space. Their means equal $\theta_{\tilde{a}}$ and $\theta_{\tilde{b}}$ respectively, so

$$\mathrm{Cov}\left[\tilde{a},\tilde{b}\right] = \mathrm{E}\left[\tilde{a}\tilde{b}\right] - \theta_{\tilde{a}}\theta_{\tilde{b}}. \tag{53}$$

Since

$$\begin{aligned}
\mathrm{E}\left[\tilde{a}\tilde{b}\right] &= \sum_{a\in\{0,1\}}\sum_{b\in\{0,1\}} ab\, p_{\tilde{a},\tilde{b}}(a,b) && (54)\\
&= p_{\tilde{a},\tilde{b}}(1,1), && (55)
\end{aligned}$$

so

$$p_{\tilde{a},\tilde{b}}(1,1) = \theta_{\tilde{a}}\theta_{\tilde{b}} = p_{\tilde{a}}(1)p_{\tilde{b}}(1). \tag{56}$$

In addition,

$$\theta_{\tilde{a}} = p_{\tilde{a}}(1) = p_{\tilde{a},\tilde{b}}(1,0) + p_{\tilde{a},\tilde{b}}(1,1) \tag{57}$$

$$= p_{\tilde{a},\tilde{b}}(1,0) + \theta_{\tilde{a}}\theta_{\tilde{b}}, \tag{58}$$

so

$$p_{\tilde{a},\tilde{b}}(1,0) = \theta_{\tilde{a}}(1-\theta_{\tilde{b}}) = p_{\tilde{a}}(1)p_{\tilde{b}}(0) \tag{59}$$

and by the same argument $p_{\tilde{a},\tilde{b}}(0,1) = p_{\tilde{a}}(0)p_{\tilde{b}}(1)$. Finally,

$$1-\theta_{\tilde{a}} = p_{\tilde{a}}(0) = p_{\tilde{a},\tilde{b}}(0,0) + p_{\tilde{a},\tilde{b}}(0,1) \tag{60}$$

$$= p_{\tilde{a},\tilde{b}}(0,0) + (1-\theta_{\tilde{a}})\theta_{\tilde{b}}, \tag{61}$$

so

$$p_{\tilde{a},\tilde{b}}(0,0) = (1-\theta_{\tilde{a}})(1-\theta_{\tilde{b}}) = p_{\tilde{a}}(0)p_{\tilde{b}}(0). \tag{62}$$

We conclude that uncorrelated Bernoulli random variables are independent.

8. (Exam)

   a) The MMSE estimator is the conditional mean, which we estimate using the sample conditional mean, depicted by the circle markers in the graph below. The advantage of this estimator is that it is very flexible, it indicates that grades increase a lot proportionally to the number of hours up to a certain point, but then there are diminishing returns (if any at all). The main disadvantage is that it is very noisy due to insufficient data, which results in overfitting. Also it does not provide an estimate at 4, 7 or 9.

   b) The least-squares simple-linear-regression estimator that approximates the linear MMSE estimator is

$$\ell(h) = \sqrt{v_{\text{grade}}}\rho_{\text{grade,hours}}\left(\frac{h - m_{\text{hours}}}{\sqrt{v_{\text{hours}}}}\right) + m_{\text{grade}} \tag{63}$$

$$= \frac{c_{\text{grade,hours}}}{v_{\text{hours}}}h + m_{\text{grade}} - \frac{c_{\text{grade,hours}}}{v_{\text{hours}}}m_{\text{hours}}, \tag{64}$$

   where the sample means of the hours and grades equal

$$m_{\text{grade}} = \frac{85+90+30+80+90+80+60+40+90+85}{10} \tag{65}$$

$$= 73, \tag{66}$$

$$m_{\text{hours}} = \frac{5+6+2+6+5+5+3+3+6+8}{10} \tag{67}$$

$$= 4.9, \tag{68}$$

the sample variance of the hours is

$$v_{\text{hours}} = \frac{(5-4.9)^2 + (6-4.9)^2 + (2-4.9)^2 + (6-4.9)^2 + (5-4.9)^2}{9} + \frac{(5-4.9)^2 + (3-4.9)^2 + (3-4.9)^2 + (6-4.9)^2 + (9-4.9)^2}{9} \tag{69}$$

$$= 4.01, \tag{70}$$

and the sample covariance is

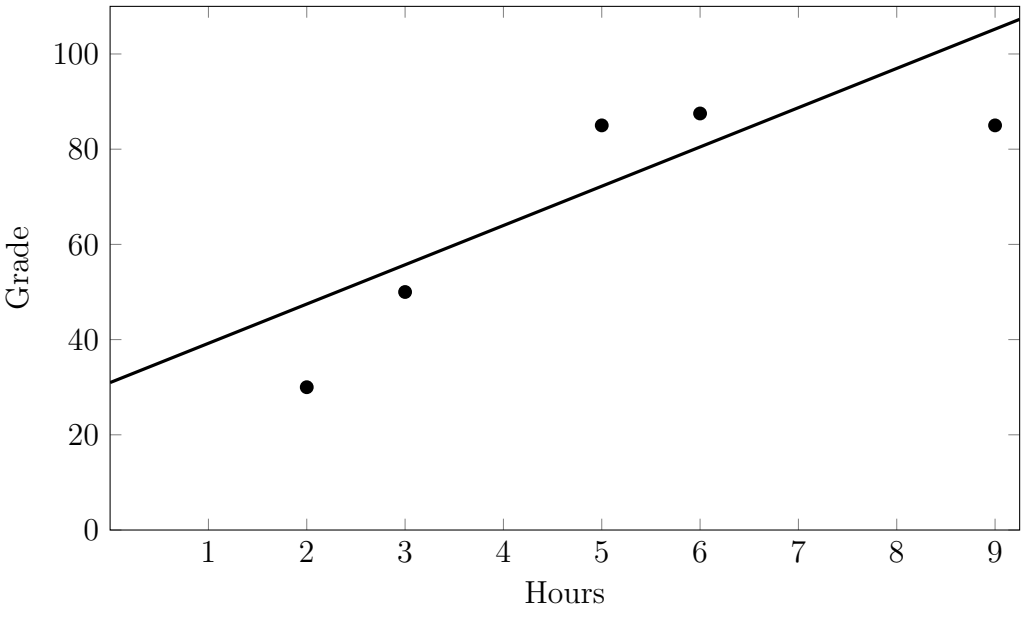
$$\begin{aligned} & c_{\text{grade,hours}} \\ &= \frac{(85-73)(5-4.9) + (90-73)(6-4.9) + (30-73)(2-4.9) + (80-73)(6-4.9)}{9} \\ &\quad + \frac{(90-73)(5-4.9) + (80-73)(5-4.9) + (60-73)(3-4.9) + (40-73)(3-4.9)}{9} \\ &\quad + \frac{(90-73)(6-4.9) + (85-73)(8-4.9)}{9} \qquad (71) \\ &= 33.11. \qquad (72) \end{aligned}$$

The least-squares estimator equals

$$\ell(h) = \frac{33.1}{4.01}h + 73 - \frac{34.4}{4.01}4.9 \tag{73}$$

$$= 8.25h + 30.97, \tag{74}$$

The estimator is depicted as a straight line on the graph below. Its advantage is that it is less noisy than the nonlinear sample conditional mean estimator, and produces estimates for any number of hours. However it underfits the data, as it does not capture the fact that the grade increase eventually saturates when the number of hours exceed 5.



9. (Online poll)

a) The proportion of Democrat voters in the whole population is $\theta_{\text{tot}} = \alpha\theta_{\text{young}} + (1-\alpha)\theta_{\text{old}}$. The fraction of young voters in the poll that intend to vote Democrat y/n_1 is a reasonable estimate for θ_{young}. The fraction of old voters in the poll that intend to vote Democrat o/n_2 is a reasonable estimate for θ_{old}. The estimator for $\theta_{\text{tot}} = \alpha\theta_{\text{young}} + (1-\alpha)\theta_{\text{old}}$ therefore equals

$$e(o, y) := \frac{\alpha y}{n_1} + \frac{(1-\alpha)o}{n_2}. \tag{75}$$

b) We have $y = 50$, $n_1 = 70$, $o = 10$, $n_2 = 30$, $\alpha = 0.25$, so

$$e(10, 50) = \frac{1}{4} \cdot \frac{50}{70} + \frac{3}{4} \cdot \frac{10}{30} = 42.9\%. \tag{76}$$

c) Let us assume that the young people in the population are sampled independently and uniformly at random with replacement, and so are the old people. Let $\tilde{y}$ denote the number of young voters in the poll that intend to vote Democrat, and let $\tilde{o}$ denote the number of old voters in the poll that intend to vote Democrat. By the properties of the sample mean computed from random samples, $\tilde{y}/n_1$ and $\tilde{o}/n_2$ are unbiased estimators of θ_{young} and θ_{old} respectively. Consequently, by linearity of expectation

$$\begin{aligned} \mathrm{E}\left[e(\tilde{o}, \tilde{y})\right] &:= \mathrm{E}\left[\frac{\alpha\tilde{y}}{n_1} + \frac{(1-\alpha)\tilde{o}}{n_2}\right] && (77)\\ &= \frac{\alpha\mathrm{E}\left[\tilde{y}\right]}{n_1} + \frac{(1-\alpha)\mathrm{E}\left[\tilde{o}\right]}{n_2} && (78)\\ &= \alpha\theta_{\text{young}} + (1-\alpha)\theta_{\text{old}} = \theta_{\text{tot}}, && (79) \end{aligned}$$

so the estimator is unbiased (under our assumptions).

d) Assuming independent and uniform random sampling with replacement,

$$\begin{aligned} \mathrm{Var}\left[\frac{\tilde{y}}{n_1}\right] &= \frac{\theta_{\text{young}}(1-\theta_{\text{young}})}{n_1}, && (80)\\ \mathrm{Var}\left[\frac{\tilde{o}}{n_2}\right] &= \frac{\theta_{\text{old}}(1-\theta_{\text{old}})}{n_2}. && (81) \end{aligned}$$

Assuming $\tilde{y}$ and $\tilde{o}$ are independent, the variance of our estimator is

$$\begin{aligned} \mathrm{Var}\left[e(\tilde{o}, \tilde{y})\right] &= \mathrm{Var}\left[\frac{\alpha\tilde{y}}{n_1} + \frac{(1-\alpha)\tilde{o}}{n_2}\right] && (82)\\ &= \alpha^2\mathrm{Var}\left[\frac{\tilde{y}}{n_1}\right] + (1-\alpha)^2\mathrm{Var}\left[\frac{\tilde{o}}{n_2}\right] && (83)\\ &= \frac{\alpha^2\theta_{\text{young}}(1-\theta_{\text{young}})}{n_1} + \frac{(1-\alpha)^2\theta_{\text{old}}(1-\theta_{\text{old}})}{n_2}. && (84) \end{aligned}$$

Since the estimator is unbiased, by Chebyshev's inequality, for any ϵ

$$\mathrm{P}\left(|e(\tilde{o},\tilde{y}) - \theta_{\text{tot}}| > \epsilon\right) \leq \frac{\mathrm{Var}\left[e(\tilde{o},\tilde{y})\right]}{\epsilon^2}, \tag{85}$$

which converges to zero as $n_1 \to \infty$ and $n_2 \to \infty$, so the estimator converges to θ_{tot} in probability and is therefore consistent.

10. (Herding)

a) Note that

$$\tilde{h}_3 = \frac{\tilde{m}_1}{4} + \frac{\frac{\tilde{m}_1+\tilde{m}_2}{2}}{4} + \frac{\tilde{m}_3}{2} \tag{86}$$

$$= \frac{3\tilde{m}_1}{8} + \frac{\tilde{m}_2}{8} + \frac{\tilde{m}_3}{2}. \tag{87}$$

Let θ_{pop} denote the population proportion, such that

$$\mathrm{E}\left[\tilde{m}_1\right] = \mathrm{E}\left[\tilde{m}_2\right] = \mathrm{E}\left[\tilde{m}_3\right] = \theta_{\text{pop}}. \tag{88}$$

By linearity of expectation and (87)

$$\mathrm{E}\left[\tilde{h}_2\right] = \frac{\mathrm{E}\left[\tilde{m}_1\right] + \mathrm{E}\left[\tilde{m}_2\right]}{2} \tag{89}$$

$$= \frac{\theta_{\text{pop}}}{2}, \tag{90}$$

$$\mathrm{E}\left[\tilde{h}_3\right] = \frac{3\mathrm{E}\left[\tilde{m}_1\right]}{8} + \frac{\mathrm{E}\left[\tilde{m}_2\right]}{8} + \frac{4\mathrm{E}\left[\tilde{m}_3\right]}{8} \tag{91}$$

$$= \theta_{\text{pop}}. \tag{92}$$

Both herded estimators are unbiased. In this situation, herding does not produce bias.

b) The polls are independent, so by (87)

$$\mathrm{Var}\left[\tilde{h}_2\right] = \frac{\mathrm{Var}\left[\tilde{m}_1\right] + \mathrm{Var}\left[\tilde{m}_2\right]}{4} \tag{93}$$

$$= \frac{s^2}{2}, \tag{94}$$

$$\mathrm{Var}\left[\tilde{h}_3\right] = \frac{9\mathrm{Var}\left[\tilde{m}_1\right]}{64} + \frac{\mathrm{Var}\left[\tilde{m}_2\right]}{64} + \frac{\mathrm{Var}\left[\tilde{m}_3\right]}{4} \tag{95}$$

$$= \frac{13s^2}{32}. \tag{96}$$

Consequently the standard errors equal

$$\mathrm{se}\left[\tilde{h}_2\right] = \frac{s}{\sqrt{2}} = 0.71s, \tag{97}$$

$$\mathrm{se}\left[\tilde{h}_3\right] = \sqrt{\frac{13}{32}}s = 0.64s. \tag{98}$$

Herding improves each individual estimator.

c) Note that by (87)

$$\tilde{h}_{\text{all}} = \frac{\tilde{m}_1}{3} + \frac{\frac{\tilde{m}_1+\tilde{m}_2}{2}}{3} + \frac{\tilde{h}_3}{3} \tag{99}$$

$$= \left(\frac{1}{3} + \frac{1}{6} + \frac{3}{24}\right)\tilde{m}_1 + \left(\frac{1}{6} + \frac{1}{24}\right)\tilde{m}_2 + \frac{\tilde{m}_3}{6} \tag{100}$$

$$= \frac{15\tilde{m}_3}{24} + \frac{5\tilde{m}_3}{24} + \frac{\tilde{m}_3}{6}. \tag{101}$$

By the independence assumption,

$$\text{Var}\left[\tilde{m}_{\text{all}}\right] = \frac{\text{Var}\left[\tilde{m}_1\right] + \text{Var}\left[\tilde{m}_2\right] + \text{Var}\left[\tilde{m}_2\right]}{9} \tag{102}$$

$$= \frac{s^2}{3}, \tag{103}$$

$$\text{Var}\left[\tilde{h}_{\text{all}}\right] = \frac{13^2\text{Var}\left[\tilde{m}_1\right]}{24^2} + \frac{5^2\text{Var}\left[\tilde{m}_2\right]}{24^2} + \frac{\text{Var}\left[\tilde{m}_3\right]}{36} \tag{104}$$

$$= \frac{133s^2}{288}. \tag{105}$$

Consequently the standard errors equal

$$\text{se}\left[\tilde{m}_{\text{all}}\right] = \frac{s}{\sqrt{3}} = 0.58s, \tag{106}$$

$$\text{se}\left[\tilde{h}_{\text{all}}\right] = \sqrt{\frac{133}{288}}s = 0.68s. \tag{107}$$

The herded estimator $\tilde{h}_{\text{all}}$ has a larger standard error than the non-herded estimator $\tilde{m}_{\text{all}}$, so herding is problematic because it degrades the aggregated estimator. This happens because the first poll (and to a lesser extent the second poll) are overrepresented in the herded estimators.

11. (Confidence intervals)

a) 72 out of 80 intervals contain μ. A reasonable estimate for $1-\alpha$ is therefore 0.9, so the estimate for α is 0.1.

b)
- $X = B$ and $Y = A$. The central limit theorem produces much narrower confidence intervals.
- $X = B$ and $Y = A$. More data results in narrower confidence intervals (see equation in the notes).
- $X = A$ and $Y = B$. Higher variance results in wider confidence intervals (see equation in the notes).

12. (Cholesterol)

a) Let $\tilde{x}_1,\ \tilde{x}_2,\ \ldots,\ \tilde{x}_{200}$ be samples from the men, and $\tilde{y}_1,\ \tilde{y}_2,\ \ldots,\ \tilde{y}_{100}$ the sample from the women. If each individual is selected independently with replacement, then $\text{Var}\left[\tilde{x}_i\right] = \sigma^2_{\text{men}}$ and $\text{Var}\left[\tilde{y}_i\right] = \sigma^2_{\text{women}}$. As a result, by independence

$$\text{Var}\left[\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i - \frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i\right] = \text{Var}\left[\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i\right] + \text{Var}\left[-\frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i\right] \tag{108}$$

$$= \text{Var}\left[\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i\right] + \text{Var}\left[\frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i\right] \tag{109}$$

$$= \frac{\sigma^2_{\text{men}}}{200} + \frac{\sigma^2_{\text{women}}}{100}. \tag{110}$$

Therefore the standard error is $\sqrt{\frac{\sigma^2_{\text{men}}}{200} + \frac{\sigma^2_{\text{women}}}{100}}$.

b) By the central limit theorem, if μ_{men} and μ_{women} are the population means for men and women respectively, then $\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i$ is approximately Gaussian with mean μ_{men} and variance $\frac{\sigma^2_{\text{men}}}{200}$, and $\frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i$ is approximately Gaussian with mean μ_{women} and variance $\frac{\sigma^2_{\text{women}}}{100}$. As a result, $\frac{\sigma^2_{\text{men}}}{200} - \frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i$ is approximately Gaussian with mean $\mu_{\text{men}} - \mu_{\text{women}}$ and variance $\frac{\sigma^2_{\text{men}}}{200} + \frac{\sigma^2_{\text{women}}}{100} = 1.5$. By Lemma **??** the following is a 0.95 confidence interval for $\mu_{\text{men}} - \mu_{\text{women}}$:

$$\left[20 - 1.96\sqrt{1.5}, 20 + 1.96\sqrt{1.5}\right] = [17.6, 22.4]. \tag{111}$$

13. (Modified Gaussian bootstrap confidence interval) Let τ denote the monotonic transformation and $h(X)$ the observed value of the estimator of interest. Since the transformed parameter is approximately Gaussian, we use bootstrap samples to estimate the standard error se_{bs} of the transformed estimator (for each bootstrap batch we compute the estimator and apply the transformation, then we compute the sample standard deviation of the transformed values). We can use the bootstrap standard error to construct a confidence interval for the transformed parameter:

$$[\ell, u] = \left[\tau\left(h(X)\right) - c_\alpha\,\text{se}_{\text{bs}}, \tau\left(h(X)\right) + c_\alpha\,\text{se}_{\text{bs}}\right], \tag{112}$$

where c_α is set to obtain a 1-α confidence interval under the Gaussian assumption. Now, if the transformed parameter of interest $\tau(\tilde{w})$ is in $[\ell, u]$ with probability $1-\alpha$, then the parameter of interest $\tilde{w}$ is in the interval $[\tau^{-1}(\ell), \tau^{-1}(u)]$ also with probability $1-\alpha$, because τ^{-1} is a monotonic function (the inverse of a monotonic function is monotonic). Consequently, the interval

$$\mathcal{I}^{[\tau]}_{1-\alpha} := \left[\tau^{-1}\left(\tau\left(h(X)\right) - c_\alpha\,\text{se}_{\text{bs}}\right), \tau^{-1}\left(\tau\left(h(X)\right) + c_\alpha\,\text{se}_{\text{bs}}\right)\right] \tag{113}$$

is a 1-α confidence interval for the parameter of interest.